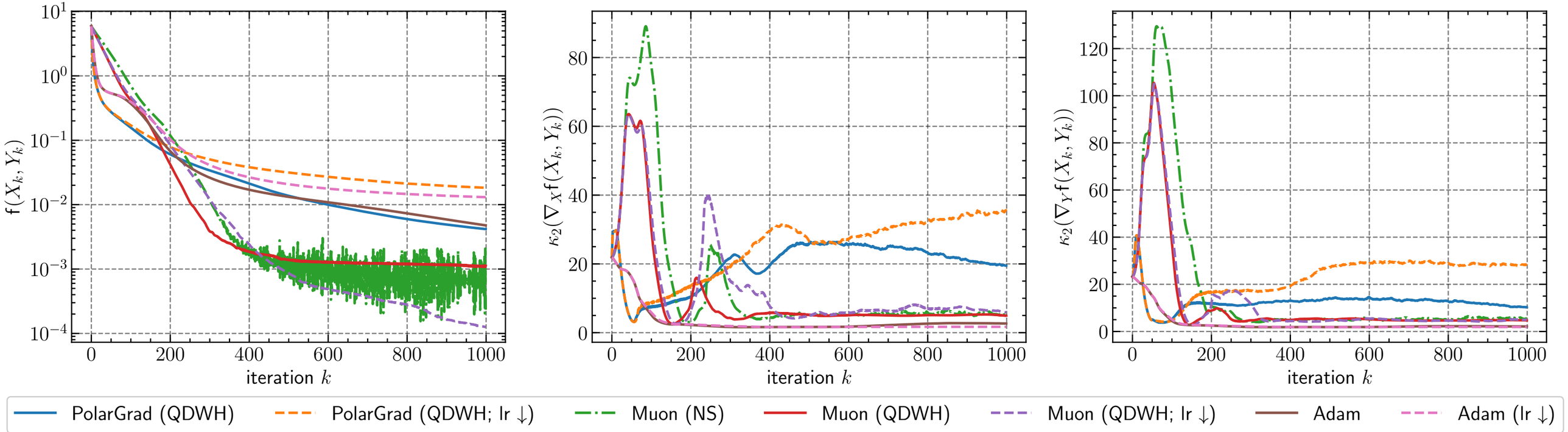
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$\mathsf{f}(X_k, Y_k)$
0
200
400
600
800
1000
iteration k
$\kappa_2(\nabla_X \mathsf{f}(X_k, Y_k))$
0
20
40
60
80
iteration k
$\kappa_2(\nabla_Y \mathsf{f}(X_k, Y_k))$
0
20
40
60
80
100
120
iteration k
PolarGrad (QDWH)
PolarGrad (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)